

DevOps Foundations

Docker & Containers - Images, containers, layers - Dockerfile (FROM, RUN, COPY, CMD, ENTRYPOINT) - Docker Compose (multi-service setups) - Container networking and volumes - Docker Hub / private registries (ECR, ACR) - Image optimization and multi-stage builds - Docker security best practices	CI/CD Pipelines - GitHub Actions (workflows, jobs, steps) - GitLab CI/CD (stages, runners) - Jenkins (pipelines, Jenkinsfile, plugins) - Pipeline concepts: build → test → deploy - Artifacts, caching, parallelism - Environment variables and secrets - Deployment strategies (blue-green, canary, rolling)
Cloud Fundamentals - Pick one: AWS / Azure / GCP (AWS recommended) - Compute: EC2, Lambda, App Service - Storage: S3, Blob Storage, GCS - Networking: VPC, subnets, security groups - Identity: IAM roles, policies, least privilege - Cloud CLI tools (aws-ali, az, gcloud) - Cost management basics	Infrastructure as Code - Terraform: providers, resources, state, modules - Terraform Cloud / remote state backends - Ansible: playbooks, roles, inventory - CloudFormation or ARM templates (cloud-native) - Drift detection and plan/apply workflow - Secrets management in IaC
Monitoring Basics - Prometheus: metrics, scraping, exporters - Grafana: dashboards, panels, alerting - CloudWatch / Azure Monitor / GCP Ops - Logs: ELK stack (Elasticsearch, Logstash, Kibana) - Alerting and on-call basics - SLI, SLO, SLA concepts	Collaboration & Culture - Agile and Scrum methodology - JIRA / Linear / Trello for ticketing - Blameless post-mortems - Runbooks and documentation - On-call rotation and incident response
TARGET CERTIFICATIONS AWS Cloud Practitioner HashiCorp Terraform Associate Docker DCA AWS Solutions Architect Associate	

- **Internal Developer Platforms**
 - Platform engineering concepts (Backstage)
 - Golden paths and self-service portals
 - Developer experience (DX) design
 - Paved roads vs freedom
 - Cost attribution and chargeback
- **SRE Practices**
 - Error budgets and burn rate alerts
 - Toil reduction automation
 - Incident management (PagerDuty, Opsgenie)
 - Runbooks and playbooks
 - Capacity planning
 - Reliability testing (load, chaos)
- **Cost Engineering (FinOps)**
 - Cloud cost visibility and tagging
 - Reserved instances vs spot vs on-demand
 - Right-sizing workloads
 - KEDA (event-driven autoscaling)
 - FinOps Foundation principles
- **Advanced Scripting & APIs**
 - Python: boto3, kubernetes client, requests
 - Go basics (many DevOps tools are in Go)
 - Webhook integrations
 - Custom Kubernetes operators (basics)
 - REST and GraphQL API design

» Cross-team incident command » Vendor evaluation and tool selection » Define SLOs and error budgets » Multi-region / multi-cloud deployments	» Enterprise security architecture » Board-level tech communication » Open-source contribution / thought leadership » Hiring and org structure design
LVL 5+ Principal / Staff Engineer	Bonus FinOps Specialist
» Defines technical direction company-wide » Writes engineering RFCs and ADRs » Works across product, infra, and security » Deep domain expertise (speaker/author level) » Drives industry standards participation	» FinOps Foundation Certified Practitioner » Cloud cost modeling and forecasting » Chargeback and showback models » Reservation and savings plan strategy » Kubernetes cost allocation (Kubecost)

DevSecOps employers – get AWS + security certs	for cloud and platform roles	unless targeting Microsoft partners	above junior level in the region
Join UAE DevOps meetups and AWS User Groups Dubai – networking opens 40–60% of senior roles	Build public GitHub projects. Hiring managers check your repos before calling you	Terraform + Kubernetes combo is the most in-demand skill pair across all UAE job listings	Add FinOps skills – UAE companies overspend on cloud and pay well for cost engineers